Course Code: CSE 2009	**Course Name: Data Communications & Computer Networks**	
Credits: 3 (2-0-2)	**Contact Hours:** 2 hour Theory & 2 hours lab per week	
Batch: 2024, 4th Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan,2026 to 8th May 2026	
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Aim of the course: This course introduces fundamental concepts in the design of computer communication networks, their protocols, and applications.

Course Overview and Context: This course provides a foundational understanding of data communication and computer networking principles essential for modern communication systems. It introduces communication models, transmission media, and physical layer concepts. The course covers data link and network layer functions such as error control, access control mechanisms, IP addressing, and routing algorithms. Transport layer services and application layer protocols enabling end-to-end communication are also discussed. The integration of theory with laboratory experiments helps students relate concepts to real-world networking scenarios.

Course Outcomes:

	Course Outcome Statement	Mapped Component	Associated Activities / Assessment
CO1	To understand the fundamental principles, architectures, and components of data communication and computer networks.	Employability	Interactive theory sessions; In-class quizzes and discussions; Demonstrations using network simulation tools
CO2	To analyze and evaluate transmission, link, and network mechanisms for reliable and efficient data delivery in computer networks.	Skill Development	Hands-on lab experiments using Cisco Packet Tracer; Analysis of ARP, IP addressing, subnetting, and routing protocols; Problem-solving exercises and lab evaluations
CO3	To apply transport and application layer protocols and services to design, configure, and assess practical networked systems and applications.	Entrepreneurship, Skill Development	Configuration of network, transport, and application layer services; Practical assessments

Topics of the course:

Topics	No. of Sessions	Course Outcome
Introduction: Data communication & components, types of data communication, Key components of computer network; network devices; types of networks; network topology, Reference models – OSI & TCP/IP	5	CO1
Physical Layer: Transmission Impairments, Transmission Medium, Transmission Modes- Half and Full Duplex	4	CO2
Data Link Layer: Error Detection and Correction- parity checks, checksum, Hamming code, Cyclic redundancy check (CRC); Flow Control Protocols: Protocols for Noiseless Channels and Noisy Channels; Ethernet- Access Control Protocols: CSMA,CSMA/CA,CSMA/CD, Token Ring- Token Passing, TDMA, FDMA, CDMA; Wireless LAN (802.11).	8	CO2
Network Layer: IP Addressing Scheme, Subnet Addressing, Subnet Masks, IPV4 Addressing, IPV6 Addressing, Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP). Unicast Routing: Routing Characteristics, Routing Algorithms: Distance Vector Routing Protocol, Link State Routing Protocol – Multicast Routing	8	CO2
Transport Layer: Services of Transport Layer, Socket Programming, TCP Phases, Transport Layer Protocols: TCP, UDP. Congestion Control Algorithms	6	CO3

Application Layer: Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP),DNS, Hypertext Transfer Protocol (HTTP), World Wide Web (WWW)	5	CO3

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1		2					3						1			
CO2				2					1	3				3		
CO3								2		3		3			2	

Learning Resources:

Textbook:

1. Behrouz A. Forouzan (2022). "Data Communications and Networking with TCPIP Protocol Suite", McGraw Hill, 6th edition.
2. Andrew S. Tanenbaum and David J. Wetherall (2013). "Computer Networks", Pearson, 5th edition.

Reference Books:

1. James F. Kurose and Keith W. Ross (2017). "Computer Networking: A Top-Down Approach", Pearson, 6th edition.
2. Larry L. Peterson and Bruce S. Davie (2021). "Computer Networks: A System Approach", Morgan Kaufmann Series, 6th edition.

Experiential Learning Component:
The Experiential Learning Component of this course is designed to strengthen students' practical understanding of data communication and computer networking concepts through hands-on laboratory work and real-world simulations. Students will perform structured experiments using tools such as Cisco Packet Tracer to design, configure, and analyze networks covering multiple layers of the OSI/TCP/IP models, including LAN setup, IP addressing and subnetting, VLANs, routing protocols, transport layer communication, and application layer services.

Assessment Pattern: The final grade will be determined by the marks or grades earned during the project's phase-wise evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics	CO mapping
Mid Term	20 %	As per academic calendar / Date-sheet	Close book written exam	CO1, CO2
Quiz	20%	March 3rd Week	Topics to be covered will be announced in the class. 20 MCQ questions, each of 1 mark. The mode of quiz will be offline.	CO1, CO2, CO3
Lab Evaluation	20%	April 4th Week	Implementation of practical learning.	CO1, CO2, CO3
End Term	40%	As per academic calendar / Date-sheet	Close book written exam	CO1, CO2, CO3

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.